

1. HIGH-LEVEL DESIGN (HLD)

1. System Overview

Problem Definition

Customer support systems often fail to provide accurate, context-aware responses when relying solely on static FAQs or generic LLM outputs. There is a need for a system that can:

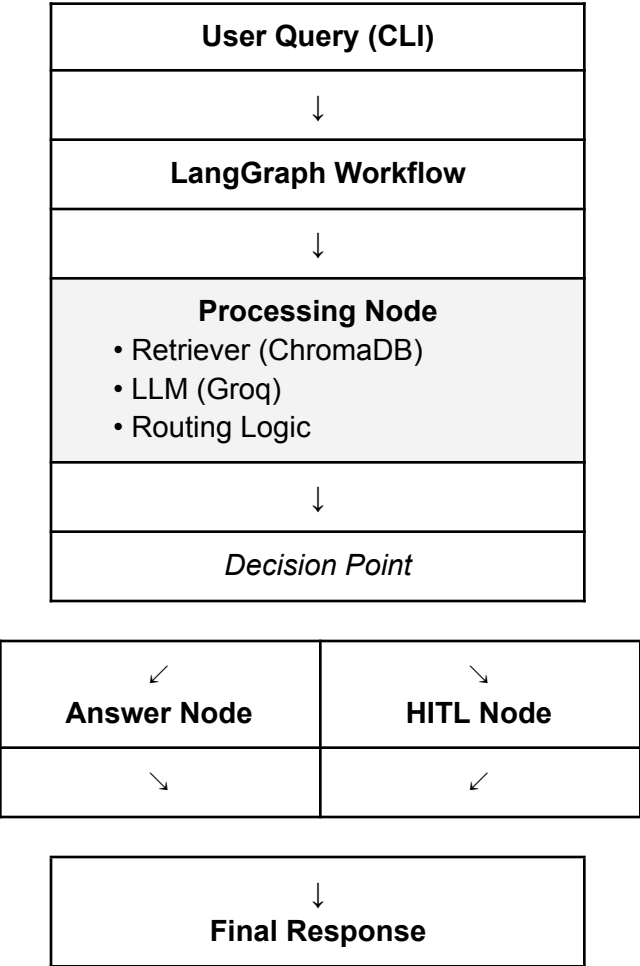
- Retrieve domain-specific knowledge
- Generate contextually accurate responses
- Escalate complex queries when needed

Scope

The system is designed to:

- Process a PDF-based knowledge base
 - Retrieve relevant information using embeddings
 - Generate responses using an LLM
 - Route queries based on intent
 - Support Human-in-the-Loop (HITL) escalation
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2. Architecture Diagram (Textual)



3. Component Description

- **Document Loader:** Loads PDF using PyPDFLoader
- **Chunking Strategy:** Splits text into chunks (400 size, 50 overlap)
- **Embedding Model:** HuggingFace embeddings
- **Vector Store:** ChromaDB for semantic retrieval
- **Retriever:** Fetches relevant chunks
- **LLM:** Groq-based model for response generation
- **Graph Workflow Engine:** LangGraph for orchestration
- **Routing Layer:** Determines answer vs escalation
- **HITL Module:** Handles unresolved queries

4. Data Flow

1. PDF → Loaded and chunked
2. Chunks → Converted to embeddings
3. Stored in ChromaDB
4. User query → Embedded
5. Relevant chunks retrieved
6. LLM generates response
7. Routing logic decides:
 - Answer OR
 - Escalation

5. Technology Choices

- **ChromaDB**: Lightweight, efficient vector storage
- **LangGraph**: Enables structured workflow and control logic
- **HuggingFace Embeddings**: Open-source, cost-effective
- **Groq LLM**: Fast inference and strong performance

6. Scalability Considerations

- Large documents → Improved chunking + indexing
- High query load → Parallel retrieval + caching
- Latency → Optimize embedding + retrieval pipeline